SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 3 – Error Analysis Task

Course Code: CSA17	**Course Title:** Artificial Intelligence
CO Assessed: CO1 – Apply (BL3)	**Assessment:** Assessment Tool 3 – Error Analysis Task
Weightage: 35% of CIA	**Duration / Marks:** 60 mins / 100 marks
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Section / Batch: _____	**Date:** 11-10-2026

CO1 Statement: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Code of Conduct. I, Deenesh Ra S T (192321149), certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Annexure A – Error Analysis Task: Answers

Conventions used throughout: an error that is merely carried forward from an earlier wrong entry is counted once, at the place where it first arises. All search traces were re-run and checked step by step.

Question 1: Foundations of AI, Intelligent Agents and Rationality

(a) Correct / incorrect statements

Stmt	Correct / Incorrect	Erroneous phrase / claim
S1	Incorrect	(i) "judges a machine to be intelligent if its internal reasoning always follows the laws of logic"; (ii) "the standard example of the 'thinking rationally' approach".
S2	Incorrect	(i) "AI winters ... occurred in the 1950s, before the 1956 Dartmouth workshop"; (ii) "That workshop then revived the field". (The claim that Dartmouth coined the term "artificial intelligence" is correct.)
S3	Incorrect	(i) "always achieves the best possible outcome because it knows the actual result of every action in advance"; (ii) "Hence ... the taxi ... was not rational" (carried forward from (i), but it is the wrong verdict on the case).
S4	Incorrect	"Performance measure: steering, accelerator, brake, horn" and "Actuators: safe, fast, legal and comfortable trip" – the P and A entries are swapped. Environment and Sensors are correct.
S5	Incorrect	"partially observable" (should be fully), "stochastic" (should be deterministic), "episodic" (should be sequential). "static" and "discrete" are correct.
S6	Correct	No erroneous phrase.

(b) Root cause analysis

Error (statement / part)	Root cause (misconception)
S1 (i)	Confuses **acting humanly** with **thinking rationally**. The Turing Test (1950) is a behavioural test: the machine passes if a human interrogator cannot tell its written answers from a human's. It says nothing about the internal reasoning obeying logic.
S1 (ii)	Mis-classifies the test among the four approaches. Thinking rationally is the "laws of thought"/logicist tradition (Aristotle, syllogisms); the Turing Test is the standard example of acting humanly.
S2 (i)	Wrong chronology. The AI winters were periods of disillusionment *after* early optimism failed: roughly mid-1970s (after the 1973 Lighthill report and the limits of early machine translation/perceptrons) and the late 1980s to early 1990s (collapse of the expert-systems market). They cannot precede the field's birth.
S2 (ii)	Mixes up cause and effect. The 1956 Dartmouth workshop *founded*/launched the field; it did not "revive" anything because nothing existed to revive.
S3 (i)	Confuses **rationality** with **omniscience**. Rationality maximises *expected* performance given what the agent knows (percepts and prior knowledge); omniscience needs knowledge of actual outcomes, which is impossible. Rationality is not perfection.
S3 (ii)	Judging an agent by the outcome rather than by the quality of its decision given the information available (outcome bias). An unforeseeable landslide cannot make a sensible decision irrational.
S4 (P and A)	Does not separate **performance measure** (how well the agent does: an external criterion of success) from **actuators** (the means by which it acts: steering, accelerator, brake, horn). The two PEAS entries were interchanged.
S5 "partially observable"	In chess the whole board is visible to the agent at every move, so the environment is fully observable. Confuses hidden opponent *intent* with hidden *state*.

Error (statement / part)	Root cause (misconception)
S5 "stochastic"	Chess has no chance element; the next state is completely determined by the current state and the action (deterministic). Unpredictability of the opponent's choice is a multi-agent issue, not stochasticity.
S5 "episodic"	A move changes the board and affects all future moves, so choices are linked across time: the task is sequential. Episodic means each percept-action episode is independent of earlier ones.

(c) Corrected statements and justification

Stmt	Corrected statement	Justification
S1	The Turing Test (1950) judges a machine intelligent if a human interrogator cannot reliably distinguish its conversational behaviour from that of a human. It is the standard example of the **acting humanly** approach.	The test is purely behavioural and judges outputs. "Thinking rationally" is the logic-based approach, which is about correct inference, not imitation of people.
S2	The "AI winters", periods of reduced funding and interest, occurred **after** the 1956 Dartmouth workshop (mid-1970s and late 1980s to early 1990s). The workshop **founded** the field and gave it the name "artificial intelligence".	Winters follow unmet expectations, so they need an existing field with early hype. Dartmouth (McCarthy, Minsky, Shannon et al.) is the birth of AI, and the term was coined there.
S3	A rational agent selects the action expected to maximise its performance measure given its percept sequence and built-in knowledge; it need not be omniscient or successful in hindsight. Hence the taxi that chose a route that every available report showed to be clear **was rational**, even though an unforeseeable landslide blocked it.	Rationality is about expected value under available information; omniscience is impossible. Information-gathering and sensible choice are what is judged, not luck.
S4	Performance measure: safe, fast, legal and comfortable trip. Environment: roads, other traffic, pedestrians, weather. Actuators: steering, accelerator, brake, horn. Sensors: cameras, GPS, speedometer.	P is the criterion used to evaluate behaviour; A are the output devices. Swapping them restores the standard PEAS description.
S5	Chess played without a clock is a **fully observable, deterministic, sequential, static, discrete** (and multi-agent) task environment.	Whole board is visible; moves have no random element; each move affects later ones; nothing changes while the agent deliberates (no clock); finite set of states and moves.
S6	No change.	Correct: a model-based reflex agent maintains internal state, updated using a model of how the world evolves independently of the agent and how its actions affect the world, so it can handle partial observability.

(d) Application: the self-driving taxi and the landslide

(i) Was the choice rational? Rationality at any time depends on four things:

Factor	Applied to the taxi
1. Performance measure that defines success	Safe, fast, legal, comfortable trip. The highway minimised expected travel time and risk.
2. Agent's prior knowledge of the environment	Knowledge of the road network and of the usual reliability of traffic reports; no knowledge of any landslide.
3. Actions the agent can perform	Choose among routes, steer, accelerate, brake. The highway was an available action.
4. Percept sequence to date	Every available traffic report showed the highway to be clear; nothing in the percepts indicated danger.

Given these four factors, choosing the highway maximised *expected* performance, so the choice **was rational**. The taxi is not omniscient, and rationality does not require success. A rational taxi would only be faulted if it ignored available percepts, e.g. took the road despite a report of a blockage.

(ii) Properties that make perfect foresight impossible: the environment is **partially observable** (the taxi cannot sense slope stability, other roads' conditions ahead, other drivers' intentions); **stochastic** (events such as landslides, accidents and sudden weather are not determined by the current state and the taxi's action); and **dynamic** (the world changes while the taxi deliberates). It is also sequential, continuous and multi-agent, which adds to the difficulty.

(iii) Simple reflex vs model-based reflex taxi. A simple reflex agent acts on the *current* percept only (condition-action rules). Example: a car overtakes and is hidden in the taxi's blind spot for a moment, or a truck momentarily hides a cyclist. In the next frame the percept shows nothing there, so the simple reflex taxi changes lane or accelerates into them. A **model-based reflex taxi** keeps an internal **state**: the last observed positions and velocities of nearby vehicles and pedestrians, updated with (1) knowledge of how the world evolves (vehicles continue moving, a hidden cyclist is probably still there) and (2) knowledge of how its own actions affect the world (steering left moves it into the adjacent lane). Using this state it can infer what it cannot currently see and behaves safely.

Question 2: Problem-Solving Agents and Problem Formulation (Missionaries and Cannibals)

(a) Correct / incorrect

Component / Note	Correct / Incorrect	What is wrong
State representation	Incorrect	(M, C) omits the **boat position**. The boat's side decides which moves are possible and is part of the state.
Initial state	Incorrect (carried forward)	(3, 3) lacks the boat side. It should be (3, 3, L). Counted with the state-representation error.
Actions	Correct	The five loads 1M, 2M, 1C, 2C, 1M + 1C are exactly the possible non-empty loads of one or two people (direction is fixed by the boat's side).
Goal test	Incorrect	M = C is also true in the initial state (3, 3) and in (2, 2), (1, 1). The goal is everybody on the right bank: (0, 0, R).
Legality check	Incorrect	Checks only the left bank. The right bank (3 - M missionaries, 3 - C cannibals) must also satisfy "no missionaries or C <= M".
Path cost	Incorrect	The objective is the fewest crossings, so each crossing costs 1, not the number of people carried.
Note N1	Incorrect	Order is wrong. Correct: goal formulation, problem formulation, search, execution.
Note N2	Incorrect	16 = 4 x 4 counts (M, C) pairs only. With the boat side there are 32 states, of which 20 are legal (10 legal (M, C) pairs x 2 boat sides).

(b) Root cause analysis

Error (component / note)	Root cause
State representation (and initial state)	A state must contain all the information needed to determine the available actions, their results and the goal. The boat location was dropped, so the state is not a complete description of the world.
Goal test	A goal test must be satisfied by exactly the goal states. M = C is a safety-style condition mistaken for the objective; it is true in non-goal states, including the start.
Legality check	A constraint ("missionaries never outnumbered") applies to **both** banks; checking one bank only treats the constraint asymmetrically and accepts states that violate it on the other bank.
Path cost	The path cost must reflect the performance measure given in the problem (fewest crossings). Cost per person is a different objective that would also favour different solutions.
N1	Violates the problem-solving agent's cycle: the agent must first formulate a goal (to limit behaviour), then formulate the problem (states and actions to consider), and only then search; execution comes after a solution has been found. One cannot search an undefined problem.
N2	Counting error from the incomplete representation: the state space size follows from the formulation, so omitting the boat halves the count. Also the 10 legal count applied to (M, C) pairs only, not to full states.

(c) Corrected formulation and justification

Component / Note	Corrected formulation	Justification
State representation	(M, C, B): M, C = missionaries and cannibals on the **left** bank (0 to 3); B in {L, R} = side of the boat.	Complete: right-bank counts are 3 - M, 3 - C, and B fixes the direction and source bank.
Initial state	(3, 3, L)	Everyone and the boat are on the left bank.
Actions	No change: carry 1M, 2M, 1C, 2C or 1M + 1C across (applicable only if the boat's bank has that many people).	Covers all legal boat loads; the boat cannot be empty or carry more than two.
Goal test	State = (0, 0, R)	Everyone, and the boat, on the right bank. Satisfied by exactly one state.
Legality check	A state is legal iff (M = 0 or C <= M) **and** (3 - M = 0 or 3 - C <= 3 - M).	Applies the safety rule to both banks; a bank with no missionaries is safe.
Path cost	1 per crossing (step cost 1).	Matches "fewest crossings"; the path cost is the number of crossings.
Note N1	Goal formulation, problem formulation, search, execution.	Correct problem-solving agent cycle.
Note N2	4 x 4 x 2 = 32 states, of which 20 are legal (10 legal (M, C) pairs for each boat side). Only 16 of the legal states are reachable from the start.	Follows from the complete state representation and the two-bank legality check (verified by enumeration and breadth-first search).

(d) Application of the corrected formulation

(i) Boat loads from the initial state (3, 3, L)

Boat load	Resulting state (M, C, boat)	Legal?	Reason
1M	(2, 3, R)	No	Left bank: 2 missionaries, 3 cannibals, so the missionaries are outnumbered.
2M	(1, 3, R)	No	Left bank: 1 missionary, 3 cannibals, so the missionary is outnumbered.
1C	(3, 2, R)	Yes	Left: 3M, 2C (safe). Right: 0M, 1C (no missionaries, safe).
2C	(3, 1, R)	Yes	Left: 3M, 1C (safe). Right: 0M, 2C (safe).
1M + 1C	(2, 2, R)	Yes	Left: 2M, 2C (equal, safe). Right: 1M, 1C (equal, safe).

(ii) A state accepted by the student's check but actually illegal: (M, C) = (2, 1), e.g. (2, 1, R). Left bank 2M, 1C passes the student's test (C <= M), but the right bank has 1 missionary and 2 cannibals, so the missionary is outnumbered. Other examples are (1, 0) (right bank 2M, 3C) and (2, 0) (right bank 1M, 3C).
(iii) Cost of an optimal solution: with 1 per crossing, the optimal solution needs **11 crossings**, so the optimal cost is **11** (confirmed by breadth-first search over the legal state space).

Question 3: Searching for Solutions and Uninformed Search

Adjacency (alphabetical): S: C, D | A: B, C | B: A, D, G | C: A, S | D: B, E, G, S | E: D, G | G: B, D, E. Edge costs: S-C 3, S-D 3, A-B 2, A-C 6, B-D 4, B-G 3, D-E 2, D-G 6, E-G 3.

(a) Errors found

No.	Where (trace / iteration / remark)	What is wrong
1	BFS, iteration 3	After expanding D the queue is written [B, E, G, A]. A was already in the queue (added at iteration 2), so the FIFO queue must be [A, B, E, G]; new children go to the back. (Carried forward: iteration 4 dequeues B instead of A, the queues in iterations 4 to 6, the missing 7th removal, "6 nodes removed".)
2	DFS, iteration 1 (repeated at iteration 2)	Children pushed in reverse alphabetical order: stack [D, C] pops D first. Alphabetically first child C must be on top: [C, D]. (Carried forward: S-D-G, cost 9, 3 removals.)
3	UCS, iteration 4	When E (g = 5) is expanded, G is reached with g = 5 + 3 = 8, cheaper than G:9(D). The entry must be replaced by G:8(E); the student kept G:9(D). (Carried forward: iteration 5 queue, expanding A at iteration 6, goal at g = 9, cost 9, 7 removals.)
4	Remark R1	False. A graph (especially an undirected one) has cycles and repeated states, so BFS does revisit nodes unless an explored set (or in-queue check) stops it. The Explored Set column is needed.
5	Remark R2, claim 1	"All three are optimal" is false: the optimal cost is 8 (S-D-E-G), not 9. BFS is optimal only for equal step costs and DFS is not optimal.
6	Remark R2, claim 2	Agreement of three results does not confirm that the traces are correct. They agree only because of errors 2 and 3; correct DFS gives cost 14 and correct UCS gives cost 8. Further, UCS must return the least-cost path, so a result of 9 is itself a warning sign.
7	Remark R3	Fewest removals is not the criterion of "best". DFS is not optimal (here cost 14, the worst) and is not complete on infinite or cyclic spaces without checks; strategies are judged by completeness, optimality, time and space.

(b) Root cause analysis

No.	Root cause (rule or concept violated)
1	BFS uses a **FIFO** queue: nodes leave in the order they entered, so the older entry A stays ahead of the new children B, E, G. The student placed A (older) at the back, i.e. broke FIFO ordering, which changes the expansion order and the final count.
2	The DFS rule was "first alphabetical child popped first". With the top of the stack on the left, pushing children so that the first child ends up leftmost gives [C, D]; the student pushed them in the order that places the last child on top (stack LIFO misunderstood).
3	UCS must keep, for every frontier node, the **cheapest** path found so far and replace entries when a cheaper path appears (relaxation). The student only inserted new nodes and ignored the better path to a node already on the frontier, so UCS stopped being optimal.
4	The role of the explored set was misunderstood: it is what makes graph search avoid repeated states and loops. Being level-by-level does not stop BFS from generating already-seen nodes (e.g. S from C or D, A from C and B).
5	Optimality conditions mixed up: BFS finds the shallowest, not cheapest, goal; DFS has no optimality guarantee; only UCS is optimal for general positive step costs. Equal results from faulty traces were treated as proof.
6	Verification logic: agreement between outputs is not validation. A trace is correct only if it obeys the stated conventions at every iteration; and an optimal-cost algorithm cannot return a cost higher than another path's cost.
7	Strategy evaluation criteria (completeness, optimality, time, space) were reduced to a single number, the nodes removed, ignoring solution quality.

(c) Corrected traces

(i) BFS (goal test when a node is dequeued; do not add explored or queued nodes)

Iteration	Node dequeued	Queue after expansion	Explored set
1	S	[C, D]	{S}
2	C	[D, A]	{S, C}
3	D	[A, B, E, G]	{S, C, D}
4	A	[B, E, G]	{S, C, D, A}
5	B	[E, G]	{S, C, D, A, B}
6	E	[G]	{S, C, D, A, B, E}
7	G (goal)	-	stop

Queue after iteration 3: [A, B, E, G]. **Order of dequeued nodes:** S, C, D, A, B, E, G.

(ii) DFS (stack, top on the left)

Iteration	Node popped	Stack after pushing children
1	S	[C, D]
2	C	[A, D] (S explored)
3	A	[B, D] (C explored)
4	B	[G, D] (A explored; D already on stack, not pushed again)
5	G (goal)	[D] - stop

(iii) UCS (entries Node:g(parent))

Iteration	Node expanded	g(n)	Priority queue after expansion
1	S	0	C:3(S), D:3(S)
2	C	3	D:3(S), A:9(C)
3	D	3	E:5(D), B:7(D), A:9(C), G:9(D)
4	E	5	B:7(D), **G:8(E)**, A:9(C) (G replaced: 8 < 9)
5	B	7	G:8(E), A:9(C) (A via B = 9 not cheaper; G via B = 10 not cheaper)
6	G (goal)	8	-

Justification of the corrections. BFS: FIFO order means A (queued first) is dequeued before B, E and G, so the goal is reached at the 7th removal, and the path is still S-D-G because G was first generated from D. DFS: with C on top the search goes down S-C-A-B, then G (parent B); D stays on the stack, unexplored, because the goal is reached first, giving S-C-A-B-G with cost 3 + 6 + 2 + 3 = 14. UCS: relaxation at iteration 4 gives G:8(E); nothing cheaper remains, so G is selected with g = 8 and the path S-D-E-G (3 + 2 + 3 = 8).

(d) Application of the corrected traces

(i) Comparison table

Strategy	Path found	Path cost	Nodes removed from frontier	Optimal here?
BFS	S-D-G	9	7	No (optimal is 8)
DFS	S-C-A-B-G	14	5	No
UCS	S-D-E-G	8	6	Yes

(ii) Why an explored set is needed. In corrected BFS iteration 3, node D is expanded and its neighbours are B, E, G *and S*. S was expanded in iteration 1 (it is in the explored set); without the explored set S would be added to the queue again [A, S, B, E, G], then expanded again and regenerate C and D, and so on. In an undirected graph every edge leads back to a parent, so without the check the search repeats states, wastes time and memory, and in DFS can loop forever. The explored set (with the in-queue check, which also prevents duplicates of A, B, G) makes each state be expanded at most once.

Question 4: Search with Partial Information – Belief-State Computation

States: S1 (L,D,D), S2 (R,D,D), S3 (L,D,C), S4 (R,D,C), S5 (L,C,D), S6 (R,C,D), S7 (L,C,C), S8 (R,C,C). Goal states: S7 and S8.

(a) Errors found

No.	Where (task / step / conclusion)	What is wrong
1	Task A, step 3 (Suck)	Belief {S6, S8} is wrong. Suck in S6 (R,C,D) cleans the right square and gives S8; Suck in S8 leaves S8. Correct belief: {S8}. (Carried forward: C1's "still contains two states".)
2	Task B, step 0 (percept update)	After [Left, Dirty] the student has {S1, S2, S3, S6}. S2 and S6 have the agent in the **Right** square and cannot give a Left percept. Correct: {S1, S3}. (Carried forward: step 1 predicted {S4, S5, S7, S8}, which should be {S5, S7}.)
3	Task B, step 2 (Right action prediction)	Predicted {S6, S7}. Right from S7 (L,C,C) moves the agent to S8 (R,C,C); S7 was left unchanged. Correct predicted belief: {S6, S8}.
4	Conclusion C1 (knowledge claim)	"the agent can never know where it is" is wrong: even in the student's own belief {S6, S8} the agent is in the Right square in both states; in the correct belief {S8} it knows everything.
5	Conclusion C1 (unsolvability claim)	"No plan can guarantee the goal ... sensorless problems cannot be solved" is false: [Suck, Right, Suck] is a solution because it ends in a belief state all of whose states are goals.
6	Conclusion C2	{S6} = (R,C,D): the agent knows its state with certainty, but S6 is **not** a goal state (right square is dirty). The goal has not been reached.
7	Conclusion C3	A belief state is a **set** (subset) of physical states, so there are up to 2^8 = 256 belief states, not 8. Belief-state search is much harder.

(b) Root cause analysis

No.	Root cause (concept or rule misapplied)
1	The transition model was not applied to every state of the belief state. The result of an action on a belief state is the **union** of results for each member: S6 + Suck goes to S8 (the right square is cleaned); S6 is not a possible result.
2	The percept update step must **filter** the predicted belief state to the states consistent with the percept. The student matched only "current square is dirty" and ignored the **location** component of [Left, Dirty], keeping S2 and S6.
3	In the prediction step each state in the belief state must be sent through the action's transition; Right changes the agent's location in every state. S7 was copied through unchanged instead of moving to S8.
4	Confuses lack of knowledge of dirt with lack of knowledge of location, and misreads the belief state: all members of {S6, S8} share location R, so location is known. A belief state describes what the agent *knows*, which can be partly certain.
5	Misses **coercion**: actions can reduce uncertainty without sensors. A sensorless problem is solved by a plan that takes the belief state into a belief state consisting only of goal states, regardless of which state the agent actually started in.
6	Confuses knowing the state with satisfying the goal. A singleton belief state means certainty; goal test depends on whether the (known) state is a goal state.
7	The size of the belief-state space was not understood: a belief state is an element of the power set of physical states (2^N), not a single state.

(c) Corrected tables and conclusions

(i) Task A

Step	Action	Belief state after action
0 (start)	-	{S1, S3, S5, S7}
1	Suck	{S5, S7} (S1 -> S5, S3 -> S7, S5 and S7 unchanged)
2	Right	{S6, S8} (S5 -> S6, S7 -> S8)
3	Suck	{S8} (S6 -> S8, S8 unchanged)

(ii) Task B

Step	Action	After action (predicted)	Percept	After percept (updated)
0 (start)	-	{S1, S2, S3, S4, S5, S6, S7, S8}	[Left, Dirty]	{S1, S3}
1	Suck	{S5, S7}	[Left, Clean]	{S5, S7}
2	Right	{S6, S8}	[Right, Dirty]	{S6}

Justification: [Left, Dirty] keeps only states with the agent on the left and the left square dirty (S1, S3). Suck sends S1 to S5 and S3 to S7. [Left, Clean] keeps both. Right sends S5 to S6 and S7 to S8. [Right, Dirty] keeps only S6 (S8 has a clean right square).

(iii) Corrected conclusions

No.	Corrected conclusion	Justification
C1	Task A: the final belief state is the singleton {S8}, a goal state. The plan [Suck, Right, Suck] reaches the goal from every initial state in {S1, S3, S5, S7}, so sensorless problems **can** be solved (by coercion), and the agent ends up knowing its state exactly.	Computed step by step above: every member of the belief state ends in S8.
C2	Task B: the belief state {S6} is a singleton, so the agent knows its location (Right) and dirt status (left clean, right dirty), but the goal has **not** been reached. It still has to perform Suck, giving S8.	S6 = (R,C,D) is not a goal state because the right square is dirty.

(d) Application of the corrected concepts

(i) Number of belief states. A belief state is any subset of the 8 physical states, so there are 2^8 = 256 possible belief states (255 if the empty set is excluded). This makes belief-state search harder because the search space grows exponentially (2^N) in the number of physical states instead of linearly (N), each node is a set needing more memory, and each action/percept update must be applied to every member.

(ii) Plan [Right, Suck] from {S1, S3, S5, S7}

Step	Action	Belief state after action
0 (start)	-	{S1, S3, S5, S7}
1	Right	{S2, S4, S6, S8} (S1 -> S2, S3 -> S4, S5 -> S6, S7 -> S8)
2	Suck	{S4, S8} (S2 -> S4, S4 -> S4, S6 -> S8, S8 -> S8)

Solution? No. The final belief state {S4, S8} contains S4 = (R,D,C), in which the left square is still dirty, so not every state in the belief state is a goal state. The plan never cleans the left square. (The plan [Suck, Right, Suck] from Task A is a solution.)

Rubrics for Error Analysis

Criteria	Weight (%)	Level 4 – Exemplary (4)	Level 3 – Proficient (3)	Level 2 – Developing (2)	Level 1 – Beginning (1)
Error Identification	20	Accurately identifies all errors with clear understanding of issues	Identifies most errors with minor omissions	Identifies limited errors; some are missed	Fails to identify key errors
Root Cause Analysis	30	Clearly explains underlying causes with strong technical reasoning	Explains causes with moderate clarity	Partial explanation; weak reasoning	Unable to identify causes of errors
Correction & Justification	25	Provides correct solutions with strong justification and logic	Provides correct corrections with some justification	Corrections partially correct or weakly justified	Incorrect or no corrections provided
Application of Concepts	15	Effectively applies relevant concepts to analyze and resolve errors	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts
Clarity & Presentation	10	Presents analysis clearly, logically, and systematically	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand

Evaluation Page (for faculty use)

Student: Deenesh Ra S T Reg. No.: 192321149

Q. No. / Criteria	Error Identification	Root Cause Analysis	Correction & Justification	Application of Concepts	Clarity & Presentation	Sub. Total
1						
2						
3						
4						
Total						
%						

Faculty In-charge

Signature: _____
Date: _____

Course Coordinator

Signature: _____
Date: _____

Program Director

Signature: _____
Date: _____